# FX-JEPA

## Learning Effect-Aware Audio Representations for Reference-Based Audio Effect Transfer

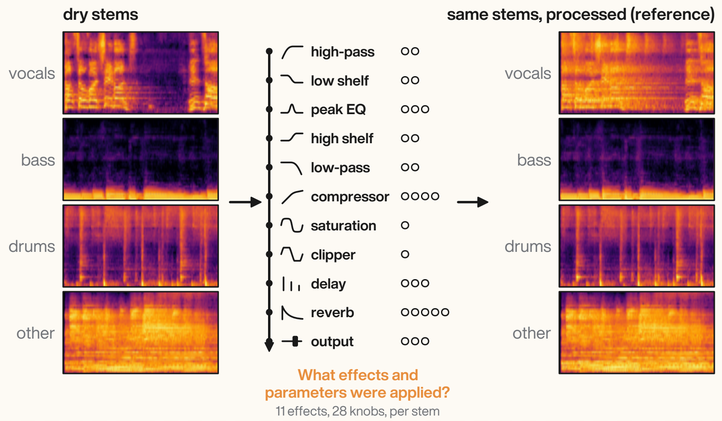
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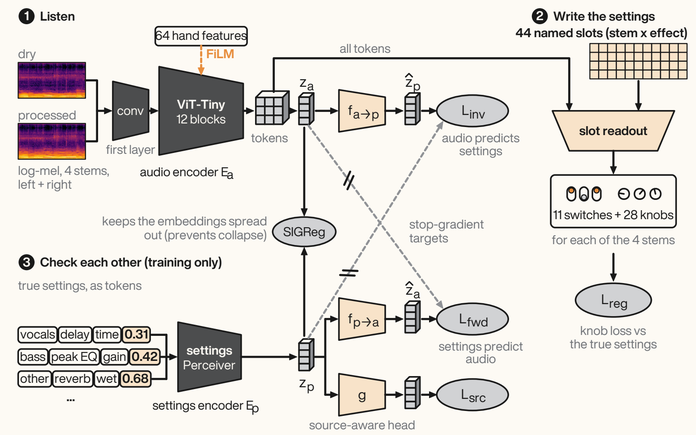
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## Reference-based effect style transfer

FX-JEPA predicts the strip settings in latent space, then renders once.

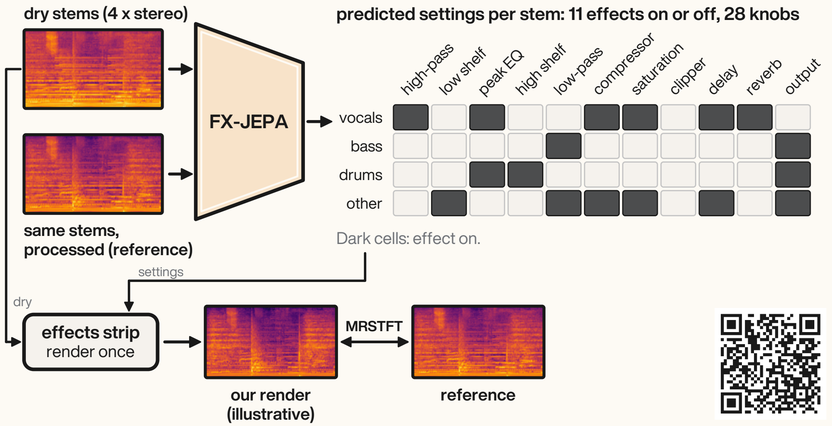



## The FX-JEPA model




$$\mathcal{L} = \frac{1-2\lambda}{3}(\mathcal{L}_{inv} + \mathcal{L}_{fwd} + \mathcal{L}_{src}) + \lambda(S_a + S_p) + w_D D + \mathcal{L}_{reg}$$

$\lambda = 0.02$, $w_D = 0.1$. $\mathcal{L}_{inv}$: audio predicts settings. $\mathcal{L}_{fwd}$: settings predict audio. $\mathcal{L}_{src}$: settings plus the dry stems predict the pair of each song. $\mathcal{L}_{reg}$: the knob loss. $S_a$, $S_p$: SIGReg, keeps each branch spread. $D$: the pairs of one song stay near their mean.






**Audio examples**
dry, reference, ours; 8 clips

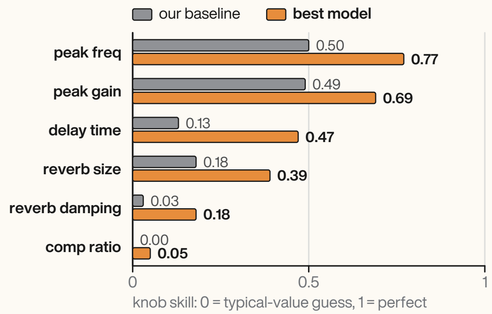
**65%**
less spectral distance than leaving the stems dry

**0.1 s**
to predict one clip's settings, on one GPU

**10.9%**
less spectral distance than our baseline, from the slot readout alone

## Ablations

The baseline predicts all 44 settings from one learned summary token. FX-JEPA gives each stem and effect its own learned query over the audio tokens. That change alone cuts the spectral distance by 10.9%, encoder unchanged.




| | median error, baseline to best |
|---|---|
| peak freq | 0.48 to 0.14 octaves |
| peak gain | 2.0 to 1.0 dB |
| delay time | 93 to 40 ms |
| reverb size | decay time 27% to 18% off |
| reverb damping | treble decay 6% to 5% off |
| comp ratio | at the guessing level |

## Spectral distance to the reference

MRSTFT, 1,000 held-out clips from 929 songs; lower is better.

**No processing**
the dry stems, unchanged — 1.17

**Best published encoder, as released**
best of four: StemFX, AFx-Rep, FX-Encoder++, RelFX; frozen, small readout on top — 0.869

**StemFX design, retrained on our data**
our data and recipe — 0.485

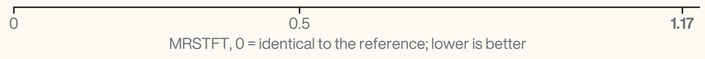
**FX-Encoder++ design, retrained on our data**
our data and recipe — 0.463

**Our baseline**
ViT-Tiny; all settings from one learned summary token; second run 0.504 — 0.492

**Baseline + conv first layer**
convolutions before the ViT; alone, within run-to-run noise — 0.495

**Baseline + slot readout**
one query per stem and effect — 0.439

**Both changes (FX-JEPA)**
conv first layer + slot readout — **0.413**




## Terms

**slot readout**
one learned query per stem and effect, 44 in all; each reads the audio tokens and predicts that effect's settings

**conv first layer**
a small convolution stack in place of the ViT's patch embedding, at the input

**JEPA**
joint embedding predictive architecture; audio and settings embeddings predict each other

**knob loss**
regression loss against the true switches and knob values

**data** FMA and MoisesDB stems; public data only; 44.1 kHz, 10 s clips

**effects** re-implemented from open-source parts: five-band EQ as in MultiAFx, compressor from torchcomp (Yu et al., 2024), tanh saturation, hard clipper, feedback delay, Freeverb reverb; fixed order